

Basic Language of Number Theory

1 Big Picture

What number theory gives us

Number theory studies integers. For algorithms, the core language is:

divisibility $\longrightarrow$ primes $\longrightarrow$ remainders $\longrightarrow$ modular classes.

Symbol	Meaning
$\mathbb{Z}$	$\{\dots, -2, -1, 0, 1, 2, \dots\}$, all integers.
$\mathbb{N}$	$\{0, 1, 2, \dots\}$, natural numbers.
d	Usually a divisor.
n	Usually a positive modulus.
q, r	Quotient and remainder.

2 Divisibility

Exact Multiple

$$d \mid a \iff a = kd \text{ for some } k \in \mathbb{Z}.$$

$$3 \mid 12$$

$$12 = 4 \cdot 3$$

no remainder

Not divide

$d \nmid a$ means d does not divide a .

Example: $5 \nmid 12$

Divides zero

Every integer divides 0.

Reason: $0 = 0 \cdot d$

Sign does not matter

$$d \mid a \iff -d \mid a$$

Example: $3 \mid 12, -3 \mid 12$

Divisor bound

If $a \neq 0$, then $1 \leq d \leq |a|$.

Example: each divisor of 24 is ≤ 24 .

Useful rules

If $d \mid a$, $d \mid b$, and $m \in \mathbb{Z}$, then

$$d \mid (a + b), \quad d \mid (a - b), \quad d \mid ma.$$

3 Divisors and Factors

Positive divisors

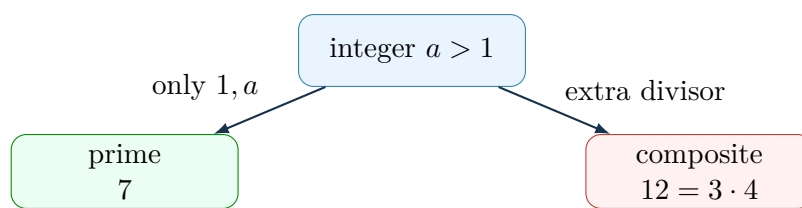
If $d \mid a$ and $d > 0$, then d is a positive divisor of a .

Integer	Positive divisors / factors
24	1, 2, 3, 4, 6, 8, 12, 24.
20	Trivial divisors: 1, 20. Nontrivial factors: 2, 4, 5, 10.

4 Prime and Composite

Classification for $a > 1$

a prime $\iff$ only positive divisors are 1, a ,
 a composite $\iff$ a has an extra positive divisor.



Type	Examples
Prime	2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
Composite	$4 = 2 \cdot 2$, $6 = 2 \cdot 3$, $9 = 3 \cdot 3$, $39 = 3 \cdot 13$.
Neither	1, 0, and negative integers.

Remember

1 is not prime. 2 is the only even prime.

5 Division Theorem

Quotient and Remainder

For $a \in \mathbb{Z}$ and $n > 0$,

$$a = qn + r, \quad 0 \leq r < n.$$

Here q is the quotient and $r = a \bmod n$ is the remainder.

Divisibility test

$$n \mid a \iff a \bmod n = 0.$$

Quick check

so

$$37 = 7 \cdot 5 + 2$$

$$37 \bmod 5 = 2.$$

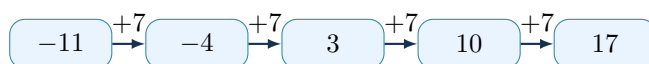
6 Modular Equivalence

Same Remainder

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Class modulo n

$$[a]_n = \{a + kn : k \in \mathbb{Z}\}.$$



$$[3]_7 = \{\dots, -11, -4, 3, 10, 17, \dots\}$$

7 $\mathbb{Z}_n$

Residue Classes

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}.$$

Short notation:

$$\mathbb{Z}_n = \{0, 1, \dots, n-1\}.$$

Each number represents a whole class. For example,

$$-1 \equiv n-1 \pmod{n},$$

so -1 represents $[n-1]_n$.

8 Final Summary

Keep These

$$d \mid a \iff a = kd, \quad k \in \mathbb{Z}.$$

$$p \text{ prime} \iff p > 1 \text{ and divisors are only } 1, p.$$

$$a = qn + r, \quad 0 \leq r < n, \quad r = a \bmod n.$$

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}.$$

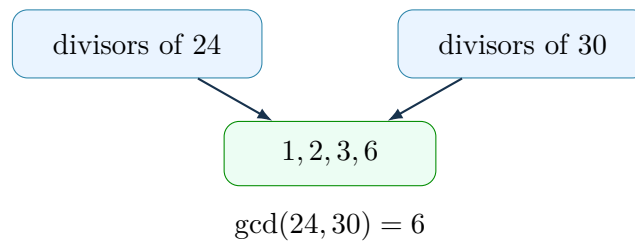
Common Divisors and GCD

1 Core Definitions

Common Divisor

An integer d is a **common divisor** of a and b when

$$d \mid a \quad \text{and} \quad d \mid b.$$



Always true

The integer 1 is a common divisor of any two integers.

2 Linear Combination Rule

Common Divisor Closure

If $d \mid a$, $d \mid b$, and $x, y \in \mathbb{Z}$, then

$$d \mid (ax + by).$$

Two useful special cases are

$$d \mid (a + b) \quad \text{and} \quad d \mid (a - b).$$

Example

Since $6 \mid 24$ and $6 \mid 30$,

$$6 \mid (30 - 24) = 6,$$

and

$$6 \mid (2 \cdot 24 - 30) = 18.$$

3 Greatest Common Divisor

Greatest Common Divisor

For integers a and b , not both zero, the **greatest common divisor** is the largest positive common divisor of a and b . It is written

$$\gcd(a, b).$$

Example	Meaning
$\gcd(24, 30) = 6$	The common divisors are 1, 2, 3, 6; the largest is 6.
$\gcd(5, 7) = 1$	The only positive common divisor is 1.
$\gcd(0, 9) = 9$	Every integer divides 0, so the largest positive divisor of 9 is 9.
$\gcd(0, 0) = 0$	Defined this way so gcd formulas work uniformly.

Range

If $a \neq 0$ and $b \neq 0$, then

$$1 \leq \gcd(a, b) \leq \min(|a|, |b|).$$

4 Basic GCD Properties

Property	Meaning
$\gcd(a, b) = \gcd(b, a)$	Order does not matter.
$\gcd(a, b) = \gcd(-a, b)$	Sign of one input does not matter.
$\gcd(a, b) = \gcd(a , b)$	GCD is controlled by absolute values.
$\gcd(a, 0) = a $	The nonzero number is the answer.
$\gcd(a, ka) = a $	A number and any integer multiple of it have gcd $ a $.

5 Bézout Characterization

Theorem

If a and b are integers, not both zero, then $\gcd(a, b)$ is the smallest positive integer of the form

$$ax + by, \quad x, y \in \mathbb{Z}.$$

The expression $ax + by$ is called an **integer linear combination** of a and b .

Proof Setup

Let

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}.$$

Since a and b are not both zero, S contains at least one positive integer. Let

$$s = \min S.$$

Then for some $x_0, y_0 \in \mathbb{Z}$,

$$s = ax_0 + by_0.$$

Step 1: Show that $s \mid a$

Divide a by s . By the division theorem, there exist integers q and r such that

$$a = qs + r, \quad 0 \leq r < s.$$

So

$$r = a - qs.$$

Substitute $s = ax_0 + by_0$:

$$r = a - q(ax_0 + by_0).$$

Distribute:

$$r = a - qax_0 - qby_0.$$

Group the a -term and the b -term:

$$r = a(1 - qx_0) + b(-qy_0).$$

Thus r is also an integer linear combination of a and b .

But $0 \leq r < s$, and s is the smallest positive linear combination. Therefore r cannot be positive. Hence

$$r = 0.$$

So

$$a = qs,$$

which means

$$s \mid a.$$

Step 2: Show that $s \mid b$

The same argument with b gives

$$b = qs + r, \quad 0 \leq r < s.$$

Using $s = ax_0 + by_0$,

$$r = b - q(ax_0 + by_0) = a(-qx_0) + b(1 - qy_0).$$

Again, r is a linear combination of a and b . Since $0 \leq r < s$, the minimality of s forces

$$r = 0.$$

Therefore,

$$\boxed{s \mid b.}$$

Conclusion so far

Since $s \mid a$ and $s \mid b$, the number s is a common divisor of a and b . Therefore,

$$\boxed{s \leq \gcd(a, b).}$$

Step 3: Show that $\gcd(a, b) \leq s$

Let

$$g = \gcd(a, b).$$

By definition,

$$g \mid a \quad \text{and} \quad g \mid b.$$

So g divides every integer linear combination:

$$g \mid (ax + by) \quad (x, y \in \mathbb{Z}).$$

In particular, since $s = ax_0 + by_0$,

$$g \mid s.$$

Because $s > 0$, this implies

$$\boxed{g \leq s.}$$

Final Result

We have

$$s \leq \gcd(a, b) \quad \text{and} \quad \gcd(a, b) \leq s.$$

Therefore,

$$\boxed{s = \gcd(a, b).}$$

So $\gcd(a, b)$ is exactly the smallest positive integer linear combination of a and b .

6 Important Corollaries

Corollary 31.3: Common divisor divides the GCD

For any integers a and b , if

$$d \mid a \quad \text{and} \quad d \mid b,$$

then

$$\boxed{d \mid \gcd(a, b)}.$$

Proof. By Bézout's characterization, there exist integers x and y such that

$$\gcd(a, b) = ax + by.$$

Since $d \mid a$ and $d \mid b$, the linear combination rule gives

$$d \mid (ax + by).$$

Therefore,

$$\boxed{d \mid \gcd(a, b)}.$$

Corollary 31.4: Scaling rule

For all integers a and b , and every nonnegative integer n ,

$$\boxed{\gcd(an, bn) = n \gcd(a, b)}.$$

Proof. If $n = 0$, then

$$\gcd(an, bn) = \gcd(0, 0) = 0$$

and

$$n \gcd(a, b) = 0 \cdot \gcd(a, b) = 0.$$

So the identity holds.

Now assume $n > 0$. By Bézout's characterization, $\gcd(an, bn)$ is the smallest positive integer of the form

$$anx + bny, \quad x, y \in \mathbb{Z}.$$

Factor out n :

$$anx + bny = n(ax + by).$$

Thus the positive linear combinations of an and bn are exactly n times the positive linear combinations of a and b . Therefore the smallest positive one is

$$n \gcd(a, b).$$

Hence

$$\boxed{\gcd(an, bn) = n \gcd(a, b)}.$$

Corollary 31.5: Cancellation rule

For all positive integers n, a, b , if

$$n \mid ab \quad \text{and} \quad \gcd(a, n) = 1,$$

then

$$\boxed{n \mid b}.$$

Proof. Since $\gcd(a, n) = 1$, Bézout's characterization gives integers x and y such that

$$ax + ny = 1.$$

Multiply both sides by b :

$$abx + nby = b.$$

Now use the assumptions:

$$n \mid ab \implies n \mid abx,$$

and

$$n \mid nby.$$

Here is why those two statements are valid:

$$n \mid ab \implies ab = nt \text{ for some } t \in \mathbb{Z}.$$

Multiplying by x ,

$$abx = ntx = n(tx),$$

so

$$n \mid abx.$$

Also,

$$nby = n(by),$$

so

$$n \mid nby.$$

Therefore n divides their sum:

$$n \mid (abx + nby).$$

But $abx + nby = b$, so

$$\boxed{n \mid b.}$$

7 Worked Example

Problem

Find $\gcd(24, 30)$, then show that every common divisor of 24 and 30 divides it.

Solution

List positive divisors:

$$\text{Div}(24) = \{1, 2, 3, 4, 6, 8, 12, 24\},$$

$$\text{Div}(30) = \{1, 2, 3, 5, 6, 10, 15, 30\}.$$

Common divisors:

$$\{1, 2, 3, 6\}.$$

Therefore,

$$\boxed{\gcd(24, 30) = 6.}$$

Each common divisor 1, 2, 3, 6 divides 6, matching the corollary.

8 Final Summary

Keep These

d common divisor of $a, b \iff d \mid a$ and $d \mid b$.

$d \mid a, d \mid b \implies d \mid (ax + by)$.

$\gcd(a, b)$ = largest positive common divisor of a, b .

$\gcd(a, b) = \min\{ax + by > 0 : x, y \in \mathbb{Z}\}$.

$d \mid a, d \mid b \implies d \mid \gcd(a, b)$.

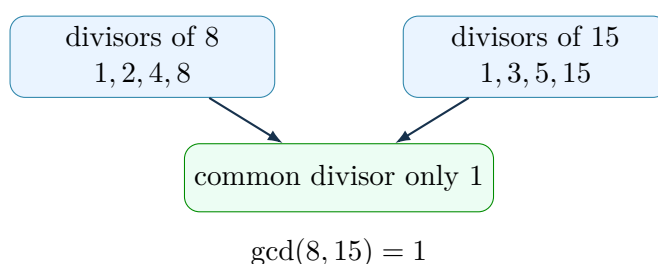
Relatively Prime Integers and Unique Factorization

1 Relatively Prime Integers

Relatively Prime

Two integers a and b are **relatively prime** if their only positive common divisor is 1. Equivalently,

$$\gcd(a, b) = 1.$$



Pair	Result
8, 15	Relatively prime, because $\gcd(8, 15) = 1$.
12, 18	Not relatively prime, because $\gcd(12, 18) = 6$.
7, 20	Relatively prime, because $\gcd(7, 20) = 1$.

2 Product Rule for Relative Primality

Theorem 31.6

For integers a , b , and p , if

$$\gcd(a, p) = 1 \quad \text{and} \quad \gcd(b, p) = 1,$$

then

$$\gcd(ab, p) = 1.$$

Proof

Since $\gcd(a, p) = 1$, Bézout's theorem gives integers x and y such that

$$ax + py = 1.$$

Since $\gcd(b, p) = 1$, there exist integers x' and y' such that

$$bx' + py' = 1.$$

Multiply the two equations:

$$(ax + py)(bx' + py') = 1.$$

Expand:

$$ab(xx') + ap(xy') + pb(yx') + p^2(yy') = 1.$$

Group every term except $ab(xx')$ as a multiple of p :

$$ab(xx') + p(axy' + byx' + pyy') = 1.$$

This writes 1 as a linear combination of ab and p . Therefore, by Bézout's theorem,

$$\gcd(ab, p) = 1.$$

Meaning

If a and b are each relatively prime to p , then their product ab is also relatively prime to p .

3 Pairwise Relatively Prime

Pairwise Relatively Prime

Integers

$$n_1, n_2, \dots, n_k$$

are **pairwise relatively prime** if every distinct pair has gcd 1:

$$i \neq j \implies \gcd(n_i, n_j) = 1.$$

Example

The integers 5, 8, 9 are pairwise relatively prime because

$$\gcd(5, 8) = 1, \quad \gcd(5, 9) = 1, \quad \gcd(8, 9) = 1.$$

Pairwise is stronger

Having $\gcd(n_1, n_2, \dots, n_k) = 1$ does not always mean the numbers are pairwise relatively prime. For example, 6, 10, 15 have no common divisor bigger than 1, but

$$\gcd(6, 10) = 2, \quad \gcd(6, 15) = 3, \quad \gcd(10, 15) = 5.$$

4 Euclid's Lemma

Theorem 31.7

For every prime p and all integers a and b , if

$$p \mid ab,$$

then

$$p \mid a \quad \text{or} \quad p \mid b \quad \text{or both.}$$

Proof

Assume

$$p \mid ab.$$

If $p \mid a$, the theorem is already true.

Now suppose

$$p \nmid a.$$

Since p is prime, its only positive divisors are 1 and p . Because $p \nmid a$, the common divisor of a and p cannot be p . Hence

$$\gcd(a, p) = 1.$$

Use the cancellation rule with $n = p$:

$$p \mid ab \quad \text{and} \quad \gcd(a, p) = 1 \implies p \mid b.$$

Therefore,

$$\boxed{p \mid a \quad \text{or} \quad p \mid b.}$$

Prime condition matters

This result is special to primes. For example,

$$6 \mid (2 \cdot 3),$$

but

$$6 \nmid 2 \quad \text{and} \quad 6 \nmid 3.$$

5 Unique Factorization

Theorem 31.8: Unique Factorization

Every composite integer a can be written in exactly one way as

$$\boxed{a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}},$$

where

$$p_1 < p_2 < \cdots < p_r$$

are primes and

$$e_1, e_2, \dots, e_r$$

are positive integers.

Meaning

Prime factorization is a unique fingerprint of a positive integer greater than 1. The order can change, but the prime bases and their exponents cannot.

Example

$$6000 = 6 \cdot 1000 = 2 \cdot 3 \cdot 10^3.$$

Since

$$10^3 = (2 \cdot 5)^3 = 2^3 \cdot 5^3,$$

we get

$$6000 = 2 \cdot 3 \cdot 2^3 \cdot 5^3.$$

Combine equal prime factors:

$$\boxed{6000 = 2^4 \cdot 3 \cdot 5^3.}$$

6 Proof of Uniqueness

Goal

Assume the same integer $n > 1$ has two prime factorizations. We prove the two lists of primes must be identical after reordering.

Write the two factorizations as products of individual primes, allowing repetition:

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where every p_i and every q_j is prime.

Step 1: Match the first prime

Since

$$p_1 \mid n,$$

and

$$n = q_1 q_2 \cdots q_s,$$

we have

$$p_1 \mid q_1 q_2 \cdots q_s.$$

By Euclid's lemma, a prime that divides a product must divide at least one factor. Therefore,

$$p_1 \mid q_j$$

for some j .

But q_j is prime. Its only positive divisors are 1 and q_j . Since $p_1 > 1$, the condition $p_1 \mid q_j$ forces

$$\boxed{p_1 = q_j.}$$

Step 2: Cancel the matched prime

Move q_j to the front of the right-hand product. Reordering multiplication does not change the product, so

$$p_1 p_2 \cdots p_r = p_1 q'_2 \cdots q'_s,$$

where $q'_2, \dots, q'_s$ are the remaining q -primes.

Cancel p_1 from both sides:

$$p_2 p_3 \cdots p_r = q'_2 \cdots q'_s.$$

Step 3: Repeat the argument

Apply the same reasoning to p_2 . Since

$$p_2 \mid q'_2 \cdots q'_s,$$

Euclid's lemma forces p_2 to equal one of the remaining q -primes. Cancel that matched prime.

Repeating this process, every prime on the left must match a prime on the right.

Step 4: The number of factors is the same

If the left side ran out first, the equality would become

$$1 = \text{a product of one or more primes},$$

which is impossible because every prime is greater than 1.

If the right side ran out first, the equality would become

$$\text{a product of one or more primes} = 1,$$

which is also impossible.

Therefore,

$$r = s.$$

Conclusion

The two factorizations contain exactly the same primes, counted with repetition. Grouping repeated equal primes gives the same prime bases and the same exponents:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \text{ uniquely.}$$

7 Final Summary

Keep These

$$\begin{aligned} \gcd(a, b) = 1 &\iff a, b \text{ are relatively prime.} \\ \gcd(a, p) = \gcd(b, p) = 1 &\implies \gcd(ab, p) = 1. \\ p \text{ prime and } p \mid ab &\implies p \mid a \text{ or } p \mid b. \\ a &= p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \text{ uniquely.} \end{aligned}$$

GCD Recursion Theorem

1 Why We Need a Better Method

Goal

We want to compute

$$\gcd(a, b)$$

efficiently for nonnegative integers a and b .

Because

$$\gcd(a, b) = \gcd(|a|, |b|),$$

it is enough to study nonnegative inputs.

Factoring is not efficient

In principle, we can compute $\gcd(a, b)$ from the prime factorizations of a and b . But factoring large integers is not known to be polynomial-time, so this is not a good algorithmic strategy.

2 GCD from Prime Factorization

Common Prime Basis

Suppose a and b use the same prime list

$$p_1, p_2, \dots, p_r,$$

with zero exponents allowed when a prime does not appear in one number.

Write

$$a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r},$$

and

$$b = p_1^{f_1} p_2^{f_2} \cdots p_r^{f_r}.$$

GCD Factorization Formula

$$\gcd(a, b) = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \cdots p_r^{\min(e_r, f_r)}.$$

The reason is simple: a common divisor can use only the prime powers that appear in both a and b . For each prime p_i , the largest exponent available to both numbers is

$$\min(e_i, f_i).$$

Example

Let

$$a = 2^4 \cdot 3^2 \cdot 5, \quad b = 2^2 \cdot 3^5 \cdot 7.$$

Using zero exponents for missing primes:

$$a = 2^4 \cdot 3^2 \cdot 5^1 \cdot 7^0,$$

$$b = 2^2 \cdot 3^5 \cdot 5^0 \cdot 7^1.$$

Therefore,

$$\gcd(a, b) = 2^{\min(4,2)} 3^{\min(2,5)} 5^{\min(1,0)} 7^{\min(0,1)}.$$

So

$$\gcd(a, b) = 2^2 \cdot 3^2 = 36.$$

3 GCD Recursion Theorem

Theorem 31.9

For every nonnegative integer a and every positive integer b ,

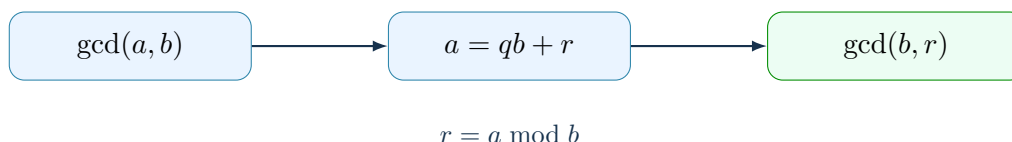
$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Why this matters

This theorem is the engine behind Euclid's algorithm. It replaces a larger pair (a, b) with a smaller pair

$$(b, a \bmod b)$$

without changing the gcd.



4 Proof of the Recursion Theorem

Setup

Let

$$r = a \bmod b.$$

By the division theorem,

$$a = qb + r, \quad 0 \leq r < b.$$

Equivalently,

$$r = a - qb.$$

Proof strategy

It is enough to show that the common divisors of a, b are exactly the same as the common divisors of b, r . Then their greatest common divisor must be the same.

Direction 1: From (a, b) to (b, r)

Assume d is a common divisor of a and b :

$$d \mid a \quad \text{and} \quad d \mid b.$$

Because $r = a - qb$, we substitute the known divisibility:

$$d \mid a \quad \text{and} \quad d \mid qb.$$

Therefore d divides the difference:

$$d \mid (a - qb).$$

But $a - qb = r$, so

$$d \mid r.$$

We already have $d \mid b$, hence

$$\boxed{d \text{ divides both } b \text{ and } r.}$$

Result of Direction 1

Every common divisor of a and b is also a common divisor of b and r .

Direction 2: From (b, r) to (a, b)

Now assume d is a common divisor of b and r :

$$d \mid b \quad \text{and} \quad d \mid r.$$

Because $a = qb + r$, we use the known divisibility:

$$d \mid qb \quad \text{and} \quad d \mid r.$$

Therefore d divides the sum:

$$d \mid (qb + r).$$

But $qb + r = a$, so

$$d \mid a.$$

We already have $d \mid b$, hence

$$\boxed{d \text{ divides both } a \text{ and } b.}$$

Result of Direction 2

Every common divisor of b and r is also a common divisor of a and b .

Conclusion

The two pairs

$$(a, b) \quad \text{and} \quad (b, r)$$

have exactly the same common divisors. Therefore their largest positive common divisor is the same:

$$\boxed{\gcd(a, b) = \gcd(b, r) = \gcd(b, a \bmod b).}$$

5 Worked Trace

Problem

Use the recursion theorem to compute $\gcd(252, 105)$.

Solution

$$252 = 2 \cdot 105 + 42,$$

so

$$\gcd(252, 105) = \gcd(105, 42).$$

Next,

$$105 = 2 \cdot 42 + 21,$$

so

$$\gcd(105, 42) = \gcd(42, 21).$$

Finally,

$$42 = 2 \cdot 21 + 0,$$

so

$$\gcd(42, 21) = 21.$$

Answer

$$\gcd(252, 105) = 21.$$

6 Final Summary

Keep These

$$\begin{aligned} a &= p_1^{e_1} \cdots p_r^{e_r}, \quad b = p_1^{f_1} \cdots p_r^{f_r} \\ \gcd(a, b) &= p_1^{\min(e_1, f_1)} \cdots p_r^{\min(e_r, f_r)} \\ a &= qb + r, \quad 0 \leq r < b, \quad r = a \bmod b \\ \gcd(a, b) &= \gcd(b, a \bmod b) \end{aligned}$$

Euclidean Algorithms

1 Goal

Problem

Given two integers a and b , compute their greatest common divisor:

$$\gcd(a, b).$$

The extended version also finds integers x and y such that

$$ax + by = \gcd(a, b).$$

The ordinary Euclidean Algorithm gives the value of the gcd. The Extended Euclidean Algorithm gives the gcd and also a linear representation of it.

2 Euclidean Algorithm

Main Identity

For integers a and $b > 0$,

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Write the division equation as

$$a = bq + r, \quad 0 \leq r < b.$$

Here $q = \lfloor a/b \rfloor$ and $r = a \bmod b$.

The reason the identity works is that every common divisor of a and b also divides

$$r = a - bq,$$

and every common divisor of b and r also divides

$$a = bq + r.$$

So the set of common divisors does not change.

3 Algorithm

Euclidean Algorithm

```
Euclid(a, b):  
  if b == 0:  
    return a  
  return Euclid(b, a mod b)
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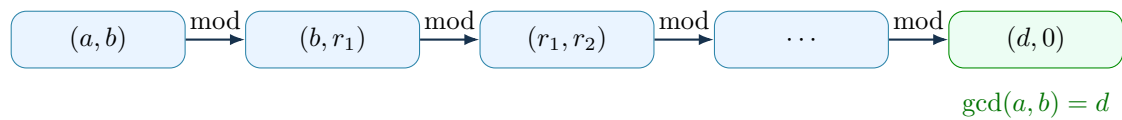
The algorithm repeatedly replaces the larger pair

$$(a, b)$$

by the smaller pair

$$(b, a \bmod b).$$

It stops when the second value becomes 0. The first value at that moment is the gcd.



Small Recursive Trace

$$\text{Euclid}(30, 21) = \text{Euclid}(21, 9) = \text{Euclid}(9, 3) = \text{Euclid}(3, 0) = 3.$$

This call sequence makes three recursive calls.

4 Worked Example: $\text{gcd}(99, 78)$

Use repeated division:

$$99 = 78(1) + 21,$$

$$78 = 21(3) + 15,$$

$$21 = 15(1) + 6,$$

$$15 = 6(2) + 3,$$

$$6 = 3(2) + 0.$$

The last nonzero remainder is 3, so

$$\boxed{\text{gcd}(99, 78) = 3.}$$

Step	Dividend	Divisor	Remainder
1	99	78	21
2	78	21	15
3	21	15	6
4	15	6	3
5	6	3	0

5 Extended Euclidean Algorithm

Bezout's Identity

For integers a and b , there exist integers x and y such that

$$\boxed{ax + by = \text{gcd}(a, b).}$$

The integers x and y are called **Bezout coefficients**. They are not always unique, but one valid pair is enough for modular inverses and many cryptographic algorithms.

6 Coefficient Recurrence

Suppose the recursive call returns

$$d = \gcd(b, a \bmod b)$$

with coefficients x_1, y_1 , so

$$bx_1 + (a \bmod b)y_1 = d.$$

Since

$$a \bmod b = a - \left\lfloor \frac{a}{b} \right\rfloor b,$$

substitute:

$$bx_1 + \left(a - \left\lfloor \frac{a}{b} \right\rfloor b \right) y_1 = d.$$

Rearrange into the form $ax + by = d$:

$$ay_1 + b \left(x_1 - \left\lfloor \frac{a}{b} \right\rfloor y_1 \right) = d.$$

Therefore,

$$x = y_1, \quad y = x_1 - \left\lfloor \frac{a}{b} \right\rfloor y_1.$$

Extended Euclidean Algorithm

```
ExtendedEuclid(a, b):  
    if b == 0:  
        return (a, 1, 0)  
  
    (d, x1, y1) = ExtendedEuclid(b, a mod b)  
  
    x = y1  
    y = x1 - floor(a / b) * y1  
  
    return (d, x, y)
```

The returned triple (d, x, y) satisfies

$$d = \gcd(a, b), \quad ax + by = d.$$

7 Back-Substitution Example

Problem

Find integers x and y such that

$$99x + 78y = \gcd(99, 78).$$

Solution

From the Euclidean Algorithm:

$$99 = 78(1) + 21, \quad 78 = 21(3) + 15,$$

$$21 = 15(1) + 6, \quad 15 = 6(2) + 3.$$

Start with the last nonzero remainder:

$$3 = 15 - 2(6).$$

Now replace each remainder using the previous division equation:

$$6 = 21 - 15.$$

So

$$3 = 15 - 2(21 - 15) = 3(15) - 2(21).$$

Next,

$$15 = 78 - 3(21),$$

hence

$$3 = 3(78 - 3(21)) - 2(21) = 3(78) - 11(21).$$

Finally,

$$21 = 99 - 78,$$

so

$$3 = 3(78) - 11(99 - 78).$$

Therefore,

$$3 = -11(99) + 14(78).$$

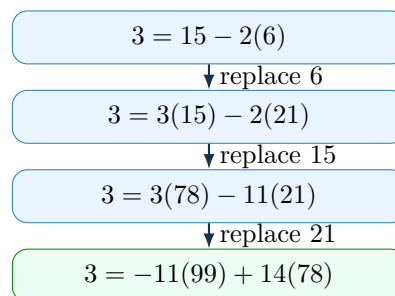
Answer

$$99(-11) + 78(14) = 3$$

Thus one valid Extended Euclidean output is

$$(d, x, y) = (3, -11, 14).$$

8 Back-Substitution Flow



9 Extended Euclid Trace Table

Trace for ExtendedEuclid(99, 78)

Back substitution is easier by hand. This table shows the same computation in the recursive form used by the algorithm.

Each row records one recursive level. The returned triple (d, x, y) satisfies

$$d = ax + by.$$

a	b	$\lfloor a/b \rfloor$	d	x	y
99	78	1	3	-11	14
78	21	3	3	3	-11
21	15	1	3	-2	3
15	6	2	3	1	-2
6	3	2	3	0	1
3	0	-	3	1	0

Returned result

The first row gives

$$(d, x, y) = (3, -11, 14),$$

so

$$99(-11) + 78(14) = 3.$$

10 Why the Algorithms Stop

In each Euclidean step,

$$a = bq + r, \quad 0 \leq r < b.$$

The second argument strictly decreases:

$$b > r \geq 0.$$

Since nonnegative integers cannot decrease forever, the process must eventually reach remainder 0.

11 Running Time

Each recursive call replaces

$$(a, b) \text{ by } (b, a \bmod b).$$

If $a < b$, the first call swaps the order:

$$\text{Euclid}(a, b) \rightarrow \text{Euclid}(b, a).$$

So we may analyze the case

$$a > b \geq 0.$$

Fibonacci Worst-Case Lemma

If $a > b \geq 1$ and $\text{Euclid}(a, b)$ performs $k \geq 1$ recursive calls, then

$$a \geq F_{k+2} \quad \text{and} \quad b \geq F_{k+1}.$$

This lemma says: many recursive calls require large inputs. The slowest cases happen when the remainders decrease like Fibonacci numbers.

Lamé's Theorem

For any integer $k \geq 1$, if

$$a > b \geq 1 \quad \text{and} \quad b < F_{k+1},$$

then $\text{Euclid}(a, b)$ makes fewer than k recursive calls.

The upper bound is tight. The call

$$\text{Euclid}(F_{k+1}, F_k)$$

makes exactly $k - 1$ recursive calls, because

$$F_{k+1} \bmod F_k = F_{k-1}.$$

Arithmetic-operation count

Since F_k grows exponentially with k , the number of recursive calls is logarithmic:

$$\text{number of Euclid calls} = \mathcal{O}(\log b).$$

Bit-operation cost

If a and b are β -bit integers, then Euclid performs $\mathcal{O}(\beta)$ arithmetic operations. With schoolbook multiplication/division costing $\mathcal{O}(\beta^2)$ bit operations, this gives

$$\mathcal{O}(\beta^3) \text{ bit operations.}$$

A tighter analysis can improve the bit bound, but this is the basic estimate from the textbook discussion.

12 Review Box

Core Facts

$\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$.
Stop when the second argument becomes 0.
Last nonzero remainder is the gcd.
 $ax + by = \text{gcd}(a, b)$ for some integers x, y .
 $x = y_1, \quad y = x_1 - \lfloor a/b \rfloor y_1$.
 $\text{Euclid}(a, b)$ makes $\mathcal{O}(\log b)$ recursive calls.

Modular Arithmetic and Groups

1 Modulo Means Remainder

Basic Idea

Modulo arithmetic keeps only the remainder.

$$17 \bmod 5 = 2$$

because

$$17 = 5 \cdot 3 + 2.$$

When we work modulo 5, every integer behaves like one of

$$0, 1, 2, 3, 4.$$

Same Remainder

$$17 \equiv 2 \pmod{5}$$

means 17 and 2 have the same remainder after division by 5.

2 Working in $\mathbb{Z}_n$

Meaning of $\mathbb{Z}_n$

$$\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$$

It is the set of possible remainders modulo n .

Example:

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}.$$

How $\mathbb{Z}_6$ is calculated

$\mathbb{Z}_6$ means: list all possible remainders after division by 6.

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}.$$

We stop at 5, because the remainder after division by 6 can never be 6.

Examples:

$$8 \bmod 6 = 2, \quad 13 \bmod 6 = 1, \quad 18 \bmod 6 = 0.$$

So 8 behaves like 2, 13 behaves like 1, and 18 behaves like 0 in $\mathbb{Z}_6$.

How to Add and Multiply

Do the normal operation first, then take the remainder.

$$4 + 5 \equiv 3 \pmod{6}$$

because

$$4 + 5 = 9, \quad 9 \bmod 6 = 3.$$

$$4 \cdot 5 \equiv 2 \pmod{6}$$

because

$$4 \cdot 5 = 20, \quad 20 \bmod 6 = 2.$$

3 Group: The Simple Meaning

Group

A **group** is a set where an operation behaves nicely:

the answer stays inside the set,
there is an identity element,
every element has an inverse,
grouping does not change the result.

Word	Simple meaning
Identity	Does nothing. For addition, identity is 0.
Inverse	Cancels an element. For addition, inverse of a is $-a$.
Abelian	Order does not matter. Example: $a + b = b + a$.
Finite	Has a limited number of elements.

4 Addition Modulo n

Always Works

$(\mathbb{Z}_n, +)$ is always a finite abelian group.

Why?

identity = 0.

The inverse of a is the number that makes the sum 0 modulo n .

Example in $\mathbb{Z}_6$

$$4 + 2 \equiv 0 \pmod{6}.$$

So the additive inverse of 4 modulo 6 is 2.

$$-4 \equiv 2 \pmod{6}.$$

5 Multiplication Modulo n

Multiplication is different from addition. Not every number has a multiplicative inverse.

Numbers With Inverses

$$\mathbb{Z}_n^* = \{a : \gcd(a, n) = 1\}$$

These are the numbers that have multiplicative inverses modulo n .

$\mathbb{Z}_n$ vs. $\mathbb{Z}_n^*$

$$\mathbb{Z}_n = \text{all remainders } 0, 1, \dots, n-1.$$

$$\mathbb{Z}_n^* = \text{only the remainders relatively prime to } n.$$

How $\mathbb{Z}_7^*$ is calculated

Start with the nonzero remainders modulo 7:

$$1, 2, 3, 4, 5, 6.$$

Keep the numbers whose gcd with 7 is 1:

$$\gcd(1, 7) = 1, \quad \gcd(2, 7) = 1, \quad \gcd(3, 7) = 1,$$

$$\gcd(4, 7) = 1, \quad \gcd(5, 7) = 1, \quad \gcd(6, 7) = 1.$$

Therefore,

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}.$$

Since 7 is prime, every nonzero remainder modulo 7 is included.

Example: $\mathbb{Z}_{15}^*$

Check the numbers from 1 to 14. Keep only the numbers relatively prime to 15:

$$\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\}.$$

Most Important Rule

$$a^{-1} \pmod{n} \text{ exists only if } \gcd(a, n) = 1.$$

If $\gcd(a, n) \neq 1$, then a has no inverse modulo n .

6 Finding a Modular Inverse

Goal

To find the inverse of a modulo n , find a number x such that

$$ax \equiv 1 \pmod{n}.$$

Then x is called the inverse of a modulo n .

Step	What to do
1	Check $\gcd(a, n) = 1$. If not, inverse does not exist.
2	Use Euclid or Extended Euclid to write $1 = ax + ny$.
3	Take the coefficient x . That is the inverse of a .
4	If x is negative, convert it to a positive residue modulo n .
5	Check by multiplying: $a \cdot x \equiv 1 \pmod{n}$.

Example

Find the inverse of 5 modulo 11.

Step 1: Check gcd

$$\gcd(5, 11) = 1.$$

So the inverse exists.

Step 2: Use Euclid

$$11 = 5(2) + 1.$$

Rewrite this to make 1 the subject:

$$1 = 11 - 5(2).$$

In Bezout form,

$$1 = 5(-2) + 11(1).$$

Step 3: Take the coefficient of 5

The coefficient of 5 is -2 . Therefore

$$5(-2) \equiv 1 \pmod{11}.$$

Step 4: Convert to a positive residue

Modulo 11,

$$-2 \equiv 9 \pmod{11}.$$

So

$$5^{-1} \equiv 9 \pmod{11}.$$

Step 5: Check

$$5 \cdot 9 = 45.$$

Since

$$45 \bmod 11 = 1,$$

So

$$5 \cdot 9 \equiv 1 \pmod{11}.$$

Answer

$$5^{-1} \equiv 9 \pmod{11}.$$

No Inverse Case

If the gcd is not 1, stop. For example, $6^{-1} \pmod{15}$ does not exist because

$$\gcd(6, 15) = 3 \neq 1.$$

7 Division Modulo n

Division Means Inverse

In modular arithmetic,

$$\frac{a}{b} \equiv a \cdot b^{-1} \pmod{n}.$$

So division by b is allowed only when b^{-1} exists.

Example:

$$\frac{4}{5} \pmod{11}$$

means

$$4 \cdot 5^{-1} \pmod{11}.$$

Since $5^{-1} \equiv 9 \pmod{11}$,

$$4 \cdot 9 = 36 \equiv 3 \pmod{11}.$$

Therefore,

$$\frac{4}{5} \equiv 3 \pmod{11}.$$

8 Euler's Phi Function

What $\phi(n)$ Counts

$\phi(n)$ = how many numbers from 1 to $n - 1$ are relatively prime to n .

Equivalently,

$$\phi(n) = |\mathbb{Z}_n^*|.$$

Phi Formula

If the different prime divisors of n are known, then

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Example: $\phi(45)$

$$45 = 3^2 \cdot 5.$$

So the different prime divisors are 3 and 5.

$$\begin{aligned}\phi(45) &= 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \\ &= 45 \cdot \frac{2}{3} \cdot \frac{4}{5} = 24.\end{aligned}$$

$$\phi(45) = 24.$$

Prime Case

If p is prime, then

$$\phi(p) = p - 1.$$

9 Subgroups

Simple Meaning

A **subgroup** is a smaller group inside a bigger group.

Example in $\mathbb{Z}_8$

The full group is

$$\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

Now take the smaller set

$$S = \{0, 2, 4, 6\}.$$

This is a subgroup because adding two numbers from S , modulo 8, always gives another number from S .

Check

$$2 + 4 = 6,$$

$$4 + 6 = 10 \equiv 2 \pmod{8},$$

$$6 + 6 = 12 \equiv 4 \pmod{8}.$$

All answers are still in

$$\{0, 2, 4, 6\}.$$

Conclusion

$\{0, 2, 4, 6\}$ is a subgroup of $\mathbb{Z}_8$.

Lagrange's Theorem

If a subgroup has size m , and the full group has size N , then

$$m \mid N.$$

Here,

$$|\mathbb{Z}_8| = 8 \quad \text{and} \quad |\{0, 2, 4, 6\}| = 4.$$

Since

$$4 \mid 8,$$

the example matches Lagrange's theorem.

10 Generated Subgroups

Simple Meaning

A **generated subgroup** starts from one element and keeps applying the group operation again and again.

generated subgroup = all elements reached from one starting element.

Example in $\mathbb{Z}_6$

Use addition modulo 6. Start with 2, then keep adding 2:

$$2,$$

$$2 + 2 = 4,$$

$$4 + 2 = 6 \equiv 0 \pmod{6},$$

$$0 + 2 = 2.$$

Now the cycle repeats:

$$2, 4, 0, 2, 4, 0, \dots$$

So

$$\langle 2 \rangle = \{0, 2, 4\}.$$

This means 2 generates $\{0, 2, 4\}$ in $\mathbb{Z}_6$.

Example in $\mathbb{Z}_7^*$

First,

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}.$$

Now start with 3 and keep multiplying by 3, always reducing modulo 7:

$$3^1 \equiv 3, \quad 3^2 \equiv 2, \quad 3^3 \equiv 6,$$

$$3^4 \equiv 4, \quad 3^5 \equiv 5, \quad 3^6 \equiv 1.$$

So

$$\langle 3 \rangle = \{1, 2, 3, 4, 5, 6\}.$$

This means 3 generates all of $\mathbb{Z}_7^*$.

Difference

subgroup = a smaller group inside a bigger group.

generated subgroup = a subgroup created by repeatedly using one element.

Every generated subgroup is a subgroup.

11 Order of an Element

Order

The **order** of an element is how many steps it takes to return to the identity.

In $\mathbb{Z}_7^*$, the identity is 1. From the previous example,

$$3^6 \equiv 1 \pmod{7}.$$

So

$$\text{ord}(3) = 6 \text{ in } \mathbb{Z}_7^*.$$

Final Exam Box

$$\mathbb{Z}_n = \{0, 1, \dots, n-1\}.$$

$(\mathbb{Z}_n, +)$ always works as a group.

$$\mathbb{Z}_n^* = \{a : \gcd(a, n) = 1\}.$$

$a^{-1} \pmod{n}$ exists iff $\gcd(a, n) = 1$.

$$\phi(n) = |\mathbb{Z}_n^*|.$$

subgroup size divides group size.

order = steps needed to return to identity.

Modular Inverses

1 Definition

Modular Inverse

An integer x is a **multiplicative inverse** of a modulo n if

$$ax \equiv 1 \pmod{n}.$$

We write

$$x \equiv a^{-1} \pmod{n}.$$

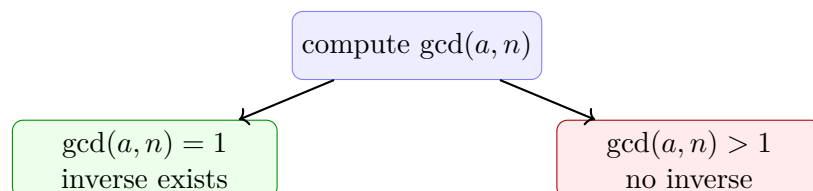
The inverse is the modular version of division. Instead of dividing by a , we multiply by a^{-1} .

2 When Does an Inverse Exist?

Existence Test

$$a^{-1} \pmod{n} \text{ exists if and only if } \gcd(a, n) = 1.$$

If a and n share a factor greater than 1, then every product ax shares that obstruction. It cannot become congruent to 1 modulo n .



3 Connection with Extended Euclid

If

$$\gcd(a, n) = 1,$$

then Bézout's identity gives integers x and y such that

$$ax + ny = 1.$$

Reduce both sides modulo n :

$$ax + ny \equiv 1 \pmod{n}.$$

Since $ny \equiv 0 \pmod{n}$,

$$ax \equiv 1 \pmod{n}.$$

Therefore

$$x \equiv a^{-1} \pmod{n}.$$

Key Point

To find $a^{-1} \pmod{n}$, run Extended Euclid on (a, n) . The coefficient of a is the inverse, reduced modulo n .

4 Algorithm

Modular Inverse by Extended Euclid

```
ModInverse(a, n):  
    (d, x, y) = ExtendedEuclid(a, n)  
  
    if d != 1:  
        return "no inverse"  
  
    return x mod n
```

The final line $x \bmod n$ is needed because Extended Euclid may return a negative coefficient.

5 Example: No Inverse

Problem

Find $2^{-1} \pmod{14}$.

Solution

Check the gcd:

$$\gcd(2, 14) = 2.$$

Since

$$2 \neq 1,$$

the inverse does not exist.

Answer

$2^{-1} \pmod{14}$ does not exist.

6 Example: Direct Reduction

Problem

Find $4^{-1} \pmod{7}$.

Solution

We need x such that

$$4x \equiv 1 \pmod{7}.$$

Try small residues:

$$4 \cdot 1 \equiv 4 \pmod{7},$$

$$4 \cdot 2 \equiv 8 \equiv 1 \pmod{7}.$$

Therefore

$$\boxed{4^{-1} \equiv 2 \pmod{7}.$$

Check

$$4 \cdot 2 = 8 \equiv 1 \pmod{7}.$$

7 Example: Extended Euclid

Problem

Find $7^{-1} \pmod{26}$.

Solution

Run Euclid:

$$26 = 7(3) + 5,$$

$$7 = 5(1) + 2,$$

$$5 = 2(2) + 1.$$

Now back-substitute:

$$1 = 5 - 2(2).$$

Since

$$2 = 7 - 5,$$

we get

$$1 = 5 - 2(7 - 5) = 3(5) - 2(7).$$

Since

$$5 = 26 - 7(3),$$

we get

$$1 = 3(26 - 7(3)) - 2(7).$$

So

$$1 = 3(26) - 11(7).$$

The coefficient of 7 is -11 . Therefore

$$7^{-1} \equiv -11 \pmod{26}.$$

Reduce:

$$-11 \equiv 15 \pmod{26}.$$

Answer

$$7^{-1} \equiv 15 \pmod{26}.$$

8 Example: Negative Coefficient

Problem

Find $5^{-1} \pmod{11}$.

Solution

Use Euclid:

$$11 = 5(2) + 1.$$

Back-substitute:

$$1 = 11 - 5(2).$$

Thus

$$1 = 5(-2) + 11(1).$$

The coefficient of 5 is -2 . Therefore

$$5^{-1} \equiv -2 \pmod{11}.$$

Reduce the negative coefficient:

$$-2 \equiv 9 \pmod{11}.$$

So

$$5^{-1} \equiv 9 \pmod{11}.$$

Always Check the Bézout Identity

The coefficient method works only when the equation is actually

$$1 = ax + ny.$$

If the right side is not 1, the coefficient is not an inverse.

9 Correct Negative Coefficient Example

Problem

Find $3^{-1} \pmod{35}$.

Solution

Use Extended Euclid:

$$35 = 3(11) + 2,$$

$$3 = 2(1) + 1.$$

Back-substitute:

$$1 = 3 - 2.$$

Since

$$2 = 35 - 3(11),$$

we get

$$1 = 3 - (35 - 3(11)).$$

Thus

$$1 = 12(3) - 35.$$

The coefficient of 3 is 12, so

$$\boxed{3^{-1} \equiv 12 \pmod{35}.$$

10 More Quick Examples

Example 1: $5x \equiv 1 \pmod{14}$

We need the inverse of 5 modulo 14.

$$\gcd(5, 14) = 1,$$

so an inverse exists.

Check small multiples:

$$5 \cdot 3 = 15 \equiv 1 \pmod{14}.$$

Therefore

$$\boxed{x \equiv 3 \pmod{14}.$$

Example 2: $35x \equiv 1 \pmod{8}$

First reduce 35 modulo 8:

$$35 \equiv 3 \pmod{8}.$$

So the equation becomes

$$3x \equiv 1 \pmod{8}.$$

Now check:

$$3 \cdot 3 = 9 \equiv 1 \pmod{8}.$$

Therefore

$$\boxed{x \equiv 3 \pmod{8}.$$

Example 3: $40x \equiv 1 \pmod{7}$

First reduce 40 modulo 7:

$$40 \equiv 5 \pmod{7}.$$

So the equation becomes

$$5x \equiv 1 \pmod{7}.$$

Now check:

$$5 \cdot 3 = 15 \equiv 1 \pmod{7}.$$

Therefore

$$x \equiv 3 \pmod{7}.$$

Example 4: $20x \equiv 1 \pmod{3}$

First reduce 20 modulo 3:

$$20 \equiv 2 \pmod{3}.$$

So the equation becomes

$$2x \equiv 1 \pmod{3}.$$

Now check:

$$2 \cdot 2 = 4 \equiv 1 \pmod{3}.$$

Therefore

$$x \equiv 2 \pmod{3}.$$

Fast Pattern

For equations of the form

$$ax \equiv 1 \pmod{n},$$

first reduce a modulo n , then find the number that multiplies it to give remainder 1.

11 Tabular Extended Euclidean Method

A common table for modular inverses tracks quotients and coefficients while running Euclid.

Step	Remainder	Quotient	Coefficient of a
0	n	—	0
1	a	—	1
2	r_2	q_1	s_2
3	r_3	q_2	s_3

When the final nonzero remainder is 1, the matching coefficient of a is the inverse modulo n .

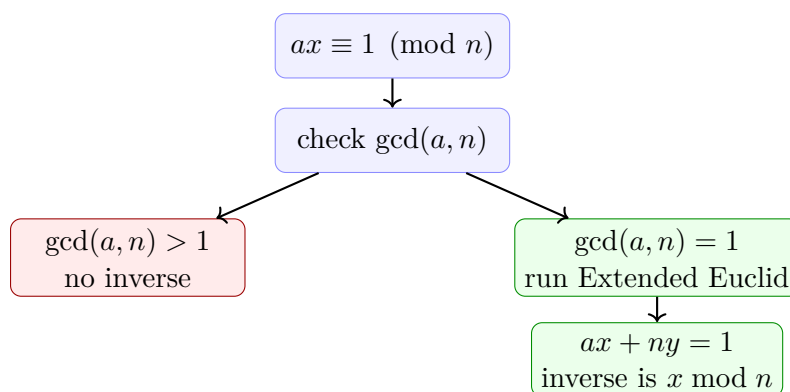
Worked Table: $3^{-1} \pmod{35}$

Step	Remainder	Equation	Coefficient of 3
0	35	35	0
1	3	3	1
2	2	$35 - 11(3)$	-11
3	1	$3 - 1(2)$	12

The coefficient of 3 when the remainder becomes 1 is 12. Hence

$$3^{-1} \equiv 12 \pmod{35}.$$

12 Visual Summary



13 Review Box

Remember

$$\begin{aligned}x &= a^{-1} \pmod{n} \iff ax \equiv 1 \pmod{n}, \\a^{-1} \pmod{n} \text{ exists} &\iff \gcd(a, n) = 1, \\ax + ny = 1 &\Rightarrow x \equiv a^{-1} \pmod{n}.\end{aligned}$$

Solving Modular Linear Equations

1 The Problem

Goal

We want to solve equations of the form

$$ax \equiv b \pmod{n}$$

where $a > 0$, $n > 0$, and x is the unknown.

This means

$$n \mid (ax - b).$$

So we are looking for all residues x modulo n that make ax have the same remainder as b .

Important

A modular linear equation may have

no solution, one solution, or more than one solution.

The gcd of a and n decides what happens.

2 The Key Number

Let

$$d = \gcd(a, n).$$

The number d measures how much a and n have in common.

Solvability Test

$$ax \equiv b \pmod{n} \text{ has a solution iff } d \mid b.$$

where

$$d = \gcd(a, n).$$

Condition	Meaning
$d \nmid b$	No solution.
$d \mid b$	Solutions exist.
$\gcd(a, n) = 1$	Exactly one solution modulo n .
$\gcd(a, n) = d > 1$ and $d \mid b$	Exactly d solutions modulo n .

3 Why the GCD Test Works

Generated Multiples

In $\mathbb{Z}_n$, the values

$$0, a, 2a, 3a, \dots$$

eventually repeat. The set generated by a is exactly

$$\langle a \rangle = \{0, d, 2d, \dots, (n/d - 1)d\}$$

where

$$d = \gcd(a, n).$$

So $ax \equiv b \pmod{n}$ is solvable exactly when b is one of these multiples of d . That is why

$$d \mid b$$

is the test.

4 How Many Solutions?

Number of Solutions

Let

$$d = \gcd(a, n).$$

Then

$$ax \equiv b \pmod{n} \text{ has either no solution or exactly } d \text{ solutions.}$$

If $d \nmid b$, there are no solutions.

If $d \mid b$, there are d distinct solutions modulo n .

5 Finding One Solution

Use Extended Euclid on a and n :

$$\text{ExtendedEuclid}(a, n) = (d, x', y').$$

This means

$$d = ax' + ny'.$$

So

$$ax' \equiv d \pmod{n}.$$

If $d \mid b$, multiply by b/d :

$$a \left(x' \frac{b}{d} \right) \equiv b \pmod{n}.$$

One Solution

$$x_0 = x' \left(\frac{b}{d} \right) \bmod n.$$

6 Finding All Solutions

Once we have one solution x_0 , the full set of solutions is:

All Solutions

$$x_i = x_0 + i \left(\frac{n}{d} \right) \pmod{n}$$

for

$$i = 0, 1, \dots, d - 1.$$

The spacing between consecutive solutions is

$$\frac{n}{d}.$$

7 Algorithm

Modular Linear Equation Solver

```
ModularLinearEquationSolver(a, b, n):  
    (d, x_prime, y_prime) = ExtendedEuclid(a, n)  
  
    if d divides b:  
        x0 = x_prime * (b / d) mod n  
  
        for i = 0 to d - 1:  
            print (x0 + i * (n / d)) mod n  
    else:  
        print "no solutions"
```

Algorithm Meaning

First find $d = \gcd(a, n)$. If d does not divide b , stop. If d divides b , compute one solution x_0 , then add n/d repeatedly to get all d solutions.

8 Worked Example

Problem

Solve

$$14x \equiv 30 \pmod{100}.$$

Step 1: Compute the gcd

$$d = \gcd(14, 100) = 2.$$

Since

$$2 \mid 30,$$

solutions exist. There will be exactly

$$\boxed{2}$$

solutions modulo 100.

Step 2: Run Extended Euclid

First run the Euclidean algorithm:

$$100 = 14(7) + 2,$$

$$14 = 2(7) + 0.$$

So

$$\gcd(14, 100) = 2.$$

Now back-substitute to write 2 as a combination of 14 and 100. From

$$100 = 14(7) + 2,$$

we get

$$2 = 100 - 14(7).$$

Rewrite this in the form

$$2 = 14x' + 100y'.$$

So

$$2 = 14(-7) + 100(1).$$

So

$$x' = -7.$$

This is the coefficient of 14, and it is the value used in the formula for x_0 .

Step 3: Compute one solution

Use

$$x_0 = x' \left(\frac{b}{d} \right) \pmod{n}.$$

Here,

$$a = 14, \quad b = 30, \quad n = 100, \quad d = 2.$$

Therefore,

$$x_0 = (-7) \left(\frac{30}{2} \right) \bmod 100.$$

$$x_0 = (-7)(15) \bmod 100.$$

$$x_0 = -105 \bmod 100.$$

$$\boxed{x_0 = 95.}$$

Step 4: Find all solutions

The spacing is

$$\frac{n}{d} = \frac{100}{2} = 50.$$

Since $d = 2$, use $i = 0, 1$:

$$x_0 + 0(50) = 95,$$

$$x_0 + 1(50) = 145 \equiv 45 \pmod{100}.$$

Answer

$$\boxed{x \equiv 95 \pmod{100}} \quad \text{or} \quad \boxed{x \equiv 45 \pmod{100}}.$$

Check

For $x = 95$:

$$14(95) = 1330 \equiv 30 \pmod{100}.$$

For $x = 45$:

$$14(45) = 630 \equiv 30 \pmod{100}.$$

So both answers are correct.

9 Special Case: Modular Inverse

When $b = 1$

The equation

$$ax \equiv 1 \pmod{n}$$

asks for the multiplicative inverse of a modulo n .

Inverse Rule

$$\boxed{a^{-1} \pmod{n} \text{ exists iff } \gcd(a, n) = 1.}$$

If it exists, the solution is unique modulo n .

10 Final Review

Core Facts

$$d = \gcd(a, n).$$

$ax \equiv b \pmod{n}$ is solvable iff $d \mid b$.

If solvable, there are exactly d solutions modulo n .

$x_0 = x'(b/d) \pmod{n}$, where $d = ax' + ny'$.

$x_i = x_0 + i(n/d) \pmod{n}, \quad i = 0, \dots, d-1.$

Chinese Remainder Theorem

1 Main Problem

Goal

The Chinese Remainder Theorem helps solve several remainder conditions at the same time:

$$\begin{aligned}x &\equiv a_1 \pmod{n_1}, \\x &\equiv a_2 \pmod{n_2}, \\&\vdots \\x &\equiv a_k \pmod{n_k}.\end{aligned}$$

The moduli must be pairwise relatively prime:

$$\gcd(n_i, n_j) = 1 \quad \text{whenever } i \neq j.$$

Main Idea

If the moduli are pairwise relatively prime, then the system has exactly one answer modulo

$$N = n_1 n_2 \cdots n_k.$$

If x_0 is the smallest nonnegative answer, then all integer answers are

$$x = x_0 + tN, \quad t \in \mathbb{Z}.$$

2 Theorem

Chinese Remainder Theorem

Suppose $n_1, n_2, \dots, n_k$ are pairwise relatively prime.

Then the system

$$x \equiv a_i \pmod{n_i} \quad (i = 1, 2, \dots, k)$$

has a unique solution modulo

$$N = n_1 n_2 \cdots n_k.$$

3 Construction Method

How to Build the Answer

For each congruence $x \equiv a_i \pmod{n_i}$:

$$M_i = \frac{N}{n_i}.$$

Then find y_i such that

$$M_i y_i \equiv 1 \pmod{n_i}.$$

Finally compute

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_k M_k y_k \pmod{N}.$$

The number y_i is the inverse of M_i modulo n_i . This inverse exists because M_i and n_i are relatively prime.

CRT Checklist

1. Multiply the moduli: $N = n_1 n_2 \dots n_k$.
2. For each equation, compute $M_i = N / n_i$.
3. Find y_i so that $M_i y_i \equiv 1 \pmod{n_i}$.
4. Add all terms $a_i M_i y_i$.
5. Reduce the final sum modulo N .

4 Why the Construction Works

The term

$$a_i M_i y_i$$

is designed to affect only the i -th congruence.

Key Switch

For the i -th modulus:

$$M_i y_i \equiv 1 \pmod{n_i}.$$

For any other modulus n_j , where $j \neq i$:

$$M_i y_i \equiv 0 \pmod{n_j}.$$

Why does the second line happen? Since

$$M_i = \frac{N}{n_i},$$

the number M_i contains every modulus except n_i . So if $j \neq i$, then $n_j \mid M_i$, and therefore

$$M_i y_i \equiv 0 \pmod{n_j}.$$

5 Proof

Part 1: A solution exists

Define

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_k M_k y_k \pmod{N}.$$

We must show that this x satisfies every congruence.

Look at one fixed modulus, say n_i . When we reduce x modulo n_i :

$$x \equiv a_1 M_1 y_1 + \cdots + a_i M_i y_i + \cdots + a_k M_k y_k \pmod{n_i}.$$

Every term except the i -th term becomes 0, because those M_j 's contain n_i as a factor:

$$a_j M_j y_j \equiv 0 \pmod{n_i} \quad (j \neq i).$$

So only the i -th term remains:

$$x \equiv a_i M_i y_i \pmod{n_i}.$$

But y_i was chosen so that

$$M_i y_i \equiv 1 \pmod{n_i}.$$

Therefore

$$x \equiv a_i \cdot 1 \pmod{n_i}.$$

So

$$\boxed{x \equiv a_i \pmod{n_i}}.$$

Since this works for every i , the constructed x is a solution.

Part 2: The solution is unique modulo N

Suppose x and z are both solutions of the same system.

Then for every i ,

$$x \equiv a_i \pmod{n_i} \quad \text{and} \quad z \equiv a_i \pmod{n_i}.$$

So

$$x \equiv z \pmod{n_i}.$$

That means

$$n_i \mid (x - z)$$

for every i .

Because the moduli $n_1, n_2, \dots, n_k$ are pairwise relatively prime, if each n_i divides $x - z$, then their product divides $x - z$:

$$N \mid (x - z).$$

Therefore

$$\boxed{x \equiv z \pmod{N}}.$$

So there is only one solution modulo N .

6 Worked Example

Problem

Solve

$$x \equiv 2 \pmod{5}$$

and

$$x \equiv 3 \pmod{13}.$$

The moduli 5 and 13 are relatively prime, so CRT applies.

$$N = 5 \cdot 13 = 65.$$

Step 1: Compute M_1 and M_2

$$M_1 = \frac{65}{5} = 13, \quad M_2 = \frac{65}{13} = 5.$$

Step 2: Find inverses

For the first congruence, find y_1 :

$$13y_1 \equiv 1 \pmod{5}.$$

Since

$$13 \equiv 3 \pmod{5}$$

and

$$3 \cdot 2 = 6 \equiv 1 \pmod{5},$$

we get

$$y_1 = 2.$$

For the second congruence, find y_2 :

$$5y_2 \equiv 1 \pmod{13}.$$

Since

$$5 \cdot 8 = 40 \equiv 1 \pmod{13},$$

we get

$$y_2 = 8.$$

Step 3: Combine

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{65}.$$

Here $a_1 = 2$ and $a_2 = 3$, so

$$x \equiv 2(13)(2) + 3(5)(8) \pmod{65}.$$

$$x \equiv 52 + 120 \pmod{65}.$$

$$x \equiv 172 \pmod{65}.$$

$$x \equiv 42 \pmod{65}.$$

Check

$$42 \equiv 2 \pmod{5}, \quad 42 \equiv 3 \pmod{13}.$$

7 Classic Three-Modulus Example

Problem

Solve

$$x \equiv 2 \pmod{3},$$

$$x \equiv 3 \pmod{5},$$

$$x \equiv 2 \pmod{7}.$$

The moduli 3, 5, 7 are pairwise relatively prime.

$$N = 3 \cdot 5 \cdot 7 = 105.$$

Congruence	n_i	a_i	$M_i = N/n_i$
1	3	2	35
2	5	3	21
3	7	2	15

Step 1: Find inverses

$$35y_1 \equiv 1 \pmod{3}.$$

Since $35 \equiv 2 \pmod{3}$ and $2 \cdot 2 \equiv 1 \pmod{3}$,

$$y_1 = 2.$$

$$21y_2 \equiv 1 \pmod{5}.$$

Since $21 \equiv 1 \pmod{5}$,

$$y_2 = 1.$$

$$15y_3 \equiv 1 \pmod{7}.$$

Since $15 \equiv 1 \pmod{7}$,

$$y_3 = 1.$$

Step 2: Combine

$$x \equiv 2(35)(2) + 3(21)(1) + 2(15)(1) \pmod{105}.$$

$$x \equiv 140 + 63 + 30 \pmod{105}.$$

$$x \equiv 233 \pmod{105}.$$

$$x \equiv 23 \pmod{105}.$$

All integer solutions are

$$x = 23 + 105t, \quad t \in \mathbb{Z}.$$

Check

$$23 \equiv 2 \pmod{3}, \quad 23 \equiv 3 \pmod{5}, \quad 23 \equiv 2 \pmod{7}.$$

8 Another Example

Problem

Solve

$$x \equiv 4 \pmod{5}$$

and

$$x \equiv 5 \pmod{11}.$$

$$N = 5 \cdot 11 = 55.$$

$$M_1 = \frac{55}{5} = 11, \quad M_2 = \frac{55}{11} = 5.$$

Find inverses:

$$11y_1 \equiv 1 \pmod{5}.$$

Since $11 \equiv 1 \pmod{5}$, we take

$$y_1 = 1.$$

$$5y_2 \equiv 1 \pmod{11}.$$

Since $5 \cdot 9 = 45 \equiv 1 \pmod{11}$, we take

$$y_2 = 9.$$

Now combine:

$$x \equiv 4(11)(1) + 5(5)(9) \pmod{55}.$$

$$x \equiv 44 + 225 \pmod{55}.$$

$$x \equiv 269 \pmod{55}.$$

$$x \equiv 49 \pmod{55}.$$

Check:

$$49 \equiv 4 \pmod{5}, \quad 49 \equiv 5 \pmod{11}.$$

9 When CRT Can Fail

Pairwise Coprime Condition

The simple CRT theorem needs pairwise relatively prime moduli.
If the moduli share a factor, the system may have no solution.

For example,

$$x \equiv 1 \pmod{4}, \quad x \equiv 2 \pmod{6}.$$

The first congruence says x is odd. The second congruence says x is even. So the system has no solution.

10 Final Review

Core Steps

$$N = n_1 n_2 \cdots n_k.$$

$$M_i = N/n_i.$$

$$\text{Choose } y_i \text{ so that } M_i y_i \equiv 1 \pmod{n_i}.$$

$$x \equiv a_1 M_1 y_1 + \cdots + a_k M_k y_k \pmod{N}.$$

Euler's Totient Function and Euler's Theorem

1 What $\phi(n)$ Counts

Definition

For $n > 1$, Euler's totient function $\phi(n)$ counts how many integers from 1 to $n - 1$ are relatively prime to n :

$$\phi(n) = |\{r : 1 \leq r < n, \gcd(r, n) = 1\}|.$$

The numbers counted by $\phi(n)$ are the invertible residues modulo n . That means each of them has a multiplicative inverse modulo n .

Why This Matters

Euler's theorem says that if a is invertible modulo n , then a certain power of a becomes 1 modulo n . This is useful for reducing large powers.

2 First Example

Find $\phi(12)$

List the numbers from 1 to 11, then keep only the ones relatively prime to 12.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

The numbers relatively prime to 12 are

1, 5, 7, 11.

So

$$\phi(12) = 4.$$

Number	Reason
1	$\gcd(1, 12) = 1$, so count it.
5	$\gcd(5, 12) = 1$, so count it.
7	$\gcd(7, 12) = 1$, so count it.
11	$\gcd(11, 12) = 1$, so count it.

3 Formula for $\phi(n)$

Totient Product Formula

If the distinct prime divisors of n are $p_1, p_2, \dots, p_k$, then

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_k}\right).$$

Equivalently,

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Common Mistake

Use only the distinct prime divisors of n .

For example, if

$$n = 2^3 \cdot 3^2,$$

then the formula uses only 2 and 3, not repeated copies of 2 and 3.

4 How to Compute $\phi(n)$

Totient Calculation Steps

1. Factor n into primes.
2. Keep only the distinct prime divisors.
3. Start with n .
4. For each distinct prime p dividing n , multiply by $(1 - 1/p)$.
5. Simplify.

Example: $\phi(45)$

Factor:

$$45 = 3^2 \cdot 5.$$

The distinct prime divisors are 3 and 5. Therefore

$$\phi(45) = 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right).$$

Simplify:

$$\phi(45) = 45 \cdot \frac{2}{3} \cdot \frac{4}{5}.$$

$$\phi(45) = 24.$$

Answer

$$\phi(45) = 24.$$

More Examples

n	Factorization	$\phi(n)$
7	7	$\phi(7) = 7(1 - \frac{1}{7}) = 6$
9	3^2	$\phi(9) = 9(1 - \frac{1}{3}) = 6$
12	$2^2 \cdot 3$	$\phi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$
20	$2^2 \cdot 5$	$\phi(20) = 20(1 - \frac{1}{2})(1 - \frac{1}{5}) = 8$

5 Prime and Prime-Power Cases

Useful Special Cases

If p is prime, then

$$\phi(p) = p - 1.$$

If p is prime and $e \geq 1$, then

$$\phi(p^e) = p^e - p^{e-1} = p^{e-1}(p - 1).$$

Why does the prime-power formula work? Among

$$1, 2, \dots, p^e - 1,$$

the only numbers not relatively prime to p^e are multiples of p . There are p^{e-1} such multiples below p^e . So

$$\phi(p^e) = p^e - p^{e-1}.$$

6 Euler's Theorem

Euler's Theorem

If

$$\gcd(a, n) = 1,$$

then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

The condition $\gcd(a, n) = 1$ is required. Without it, a may not be invertible modulo n , and the theorem may fail.

7 Proof of Euler's Theorem

Short Proof

List all $\phi(n)$ invertible residues modulo n , and multiply each of them by a .

Since $\gcd(a, n) = 1$, multiplication by a only rearranges the invertible residues. It cannot merge two different residues, because

$$ax \equiv ay \pmod{n} \Rightarrow x \equiv y \pmod{n}.$$

So the product after multiplication is congruent to the original product:

$$a^{\phi(n)} \cdot (\text{original product}) \equiv \text{original product} \pmod{n}.$$

The original product is invertible modulo n , so we cancel it:

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

8 Fermat's Little Theorem

Prime Modulus Version

If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

This is Euler's theorem with $n = p$, because

$$\phi(p) = p - 1.$$

9 Using Euler's Theorem

Problem

Find the remainder when

$$7^{50}$$

is divided by 20.

First check:

$$\gcd(7, 20) = 1.$$

So Euler's theorem applies.

Compute

$$\phi(20) = 20 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 8.$$

Thus

$$7^8 \equiv 1 \pmod{20}.$$

Reduce the exponent:

$$50 = 8(6) + 2.$$

So

$$7^{50} = 7^{8 \cdot 6 + 2} = (7^8)^6 \cdot 7^2.$$

Modulo 20:

$$7^{50} \equiv 1^6 \cdot 49 \equiv 9 \pmod{20}.$$

Answer

$$7^{50} \equiv 9 \pmod{20}.$$

10 Another Quick Example

Problem

Evaluate

$$5^4 \pmod{12}.$$

Since

$$\gcd(5, 12) = 1$$

and

$$\phi(12) = 4,$$

Euler's theorem gives

$$5^4 \equiv 1 \pmod{12}.$$

Answer

$$5^4 \equiv 1 \pmod{12}.$$

11 Final Review

Core Facts

$\phi(n)$ counts residues $1 \leq r < n$ with $\gcd(r, n) = 1$.

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

$\phi(p) = p - 1$ for prime p .

$$\phi(p^e) = p^{e-1}(p - 1).$$

$$\gcd(a, n) = 1 \Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}.$$

Fermat's Little Theorem

1 What It Says

Fermat's Little Theorem

If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

The condition $p \nmid a$ means that a is not divisible by p . Since p is prime, this is the same as

$$\gcd(a, p) = 1.$$

Why It Matters

Fermat's Little Theorem helps reduce very large powers modulo a prime.

$$a^{p-1} \equiv 1 \pmod{p}$$

lets us replace exponents by their remainders modulo $p - 1$.

2 Needed Modular Arithmetic

Congruence

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

This means a and b leave the same remainder when divided by n .

For example,

$$35 \equiv 2 \pmod{3},$$

because

$$35 - 2 = 33$$

and

$$3 \mid 33.$$

Cancellation Rule

From

$$ac \equiv bc \pmod{n},$$

we can cancel c only when

$$\gcd(c, n) = 1.$$

For Fermat's theorem, the important fact is:

$$\phi(p) = p - 1 \quad \text{when } p \text{ is prime.}$$

3 Proof

Use Euler's Theorem

Euler's theorem says:

$$\gcd(a, n) = 1 \Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}.$$

If p is prime and $p \nmid a$, then

$$\gcd(a, p) = 1.$$

Apply Euler's theorem with

$$n = p.$$

Then

$$a^{\phi(p)} \equiv 1 \pmod{p}.$$

Since p is prime,

$$\phi(p) = p - 1.$$

Therefore,

$$a^{p-1} \equiv 1 \pmod{p}.$$

4 Equivalent Form

Another Common Version

For every integer a and prime p ,

$$a^p \equiv a \pmod{p}.$$

Why?

If $p \mid a$, then both sides are 0 modulo p .

If $p \nmid a$, then Fermat's theorem gives

$$a^{p-1} \equiv 1 \pmod{p}.$$

Multiply both sides by a :

$$a^p \equiv a \pmod{p}.$$

5 Worked Example

Problem

Verify Fermat's theorem for

$$a = 11, \quad p = 13.$$

Since 13 is prime and $13 \nmid 11$, Fermat's theorem says

$$11^{12} \equiv 1 \pmod{13}.$$

Compute using repeated squaring:

$$11^2 = 121 \equiv 4 \pmod{13}.$$

$$11^4 \equiv 4^2 = 16 \equiv 3 \pmod{13}.$$

$$11^8 \equiv 3^2 = 9 \pmod{13}.$$

Since

$$12 = 8 + 4,$$

we get

$$11^{12} = 11^8 \cdot 11^4.$$

Therefore

$$11^{12} \equiv 9 \cdot 3 = 27 \equiv 1 \pmod{13}.$$

Answer

$$11^{12} \equiv 1 \pmod{13}.$$

6 Large Power Example

Problem

Find

$$7^{100} \pmod{13}.$$

Since 13 is prime and $13 \nmid 7$,

$$7^{12} \equiv 1 \pmod{13}.$$

Reduce the exponent modulo 12:

$$100 = 12(8) + 4.$$

So

$$7^{100} = 7^{12 \cdot 8 + 4} = (7^{12})^8 \cdot 7^4.$$

Modulo 13:

$$7^{100} \equiv 1^8 \cdot 7^4 \equiv 7^4 \pmod{13}.$$

Now compute

$$7^2 = 49 \equiv 10 \pmod{13}.$$

Then

$$7^4 \equiv 10^2 = 100 \equiv 9 \pmod{13}.$$

Answer

$$7^{100} \equiv 9 \pmod{13}.$$

7 Final Review

Core Facts

p prime and $p \nmid a \Rightarrow a^{p-1} \equiv 1 \pmod{p}$.

Equivalent form: $a^p \equiv a \pmod{p}$.

Use it to reduce powers modulo a prime.

It follows from Euler's theorem since $\phi(p) = p - 1$.

Sieve of Eratosthenes

1 Problem

Goal

Given an integer n , find all prime numbers from 2 to n .

Main Idea

Start by assuming every number is prime. Then repeatedly choose the next unmarked number p , and mark all multiples of p as composite.

2 Why the Sieve Works

If a number m is composite, then

$$m = ab$$

for some integers $a, b > 1$.

At least one of a and b is at most $\sqrt{m}$. So every composite number has a prime factor at most its square root.

Stopping Rule

When sieving up to n , it is enough to mark multiples for

$$p \leq \sqrt{n}.$$

3 Algorithm Steps

1. Create a Boolean array `isPrime[0...n]`.
2. Mark 0 and 1 as not prime.
3. Initially mark every number $2, 3, \dots, n$ as prime.
4. For each $p = 2, 3, \dots, \lfloor \sqrt{n} \rfloor$, if p is still marked prime, mark

$$p^2, p^2 + p, p^2 + 2p, \dots$$

as composite.

5. The numbers still marked prime are the primes up to n .

Sieve of Eratosthenes

```
Sieve(n):  
  for i = 0 to n:  
    isPrime[i] = true
```

```

isPrime[0] = false
isPrime[1] = false

for p = 2 to floor(sqrt(n)):
    if isPrime[p] == true:
        for multiple = p*p to n step p:
            isPrime[multiple] = false

return all i such that isPrime[i] == true

```

4 Why Start from p^2 ?

Important

When processing p , multiples smaller than p^2 have already been marked by smaller prime factors.

For example, when $p = 5$, the smaller multiples are

$$10 = 2 \cdot 5, \quad 15 = 3 \cdot 5, \quad 20 = 4 \cdot 5.$$

These were already handled before reaching 5. So the first new multiple is

$$5^2 = 25.$$

5 Worked Example: Primes up to 30

Problem

Find all primes from 2 to 30 using the Sieve of Eratosthenes.

Step 1: Start

Begin with

2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30.

Step 2: Mark Multiples of 2

Keep 2. Mark

4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30

as composite.

Step 3: Mark Multiples of 3

Keep 3. Start from

$$3^2 = 9.$$

Mark

9, 12, 15, 18, 21, 24, 27, 30

as composite.

Step 4: Mark Multiples of 5

Keep 5. Start from

$$5^2 = 25.$$

Mark

25, 30

as composite.

Step 5: Stop

Since

$$\sqrt{30} < 6,$$

we only need to check primes up to 5.

Answer

The primes up to 30 are

2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

6 Running Time

The sieve marks multiples of primes:

$$\frac{n}{2} + \frac{n}{3} + \frac{n}{5} + \cdots.$$

The standard running time is

$$\mathcal{O}(n \log \log n).$$

The array uses

$$\mathcal{O}(n)$$

space.

7 Final Review

Core Facts

The sieve finds all primes up to n .
Only unmarked numbers are used as new primes.
For prime p , start marking from p^2 .
Stop after $p > \sqrt{n}$.
Time: $\mathcal{O}(n \log \log n)$, space: $\mathcal{O}(n)$.

RSA Cryptography

1 Overview

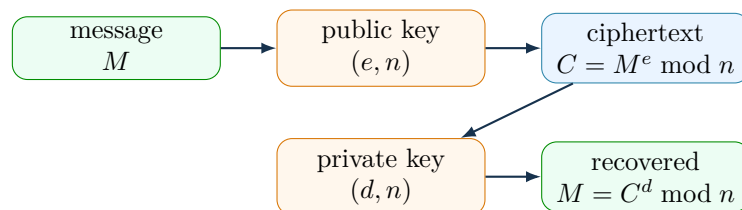
RSA

RSA is a public-key cryptosystem. It uses one key for encryption and a different related key for decryption.

public key encrypts, private key decrypts.

RSA depends on five number-theory ideas:

Idea	Role in RSA
Prime numbers	Two distinct primes p, q create the modulus $n = pq$.
Euler's totient	$\phi(n) = (p - 1)(q - 1)$ controls the exponent cycle.
Modular inverse	The private exponent d is the inverse of e modulo $\phi(n)$.
Modular exponentiation	Encryption and decryption are powers modulo n .
Factoring difficulty	Knowing n should not make it easy to recover p, q .



2 Key Generation

RSA Key Generation

1. Choose distinct primes p, q .
2. $n = pq$.
3. $\phi(n) = (p - 1)(q - 1)$.
4. Choose e with $1 < e < \phi(n)$, $\gcd(e, \phi(n)) = 1$.
5. Find d such that $de \equiv 1 \pmod{\phi(n)}$.

The public and private keys are

Public key = (e, n)

and

Private key = (d, n) .

The exponent e is public. The exponent d must remain private.

3 Encryption and Decryption

Let M be the message encoded as an integer satisfying

$$0 \leq M < n.$$

RSA Operations

$$C = M^e \bmod n$$

$$M = C^d \bmod n$$

Here C is the ciphertext. Decryption works because d was chosen to undo exponent e modulo $\phi(n)$.

4 Small RSA Example

Problem

Generate RSA keys using

$$p = 5, \quad q = 3,$$

then encrypt and decrypt

$$M = 8.$$

Solution

Compute the modulus:

$$n = pq = 5 \cdot 3 = 15.$$

Compute the totient:

$$\phi(n) = (5 - 1)(3 - 1) = 4 \cdot 2 = 8.$$

Choose

$$e = 3$$

because

$$\gcd(3, 8) = 1.$$

Now find d :

$$3d \equiv 1 \pmod{8}.$$

Since

$$3 \cdot 3 = 9 \equiv 1 \pmod{8},$$

we get

$$d = 3.$$

Thus

$$\text{Public key} = (3, 15), \quad \text{Private key} = (3, 15).$$

Encrypt:

$$C = 8^3 \bmod 15.$$

Compute

$$8^3 = 512, \quad 512 = 15(34) + 2.$$

So

$$C = 2.$$

Decrypt:

$$M = C^d \bmod 15 = 2^3 \bmod 15 = 8.$$

Answer

The original message is recovered:

$$M = 8.$$

5 Larger Key Example

Problem

Generate RSA keys using

$$p = 11, \quad q = 13,$$

and choose $e = 7$.

Solution

Compute

$$n = 11 \cdot 13 = 143.$$

Compute the totient:

$$\phi(n) = (11 - 1)(13 - 1) = 10 \cdot 12 = 120.$$

The chosen public exponent is valid because

$$\gcd(7, 120) = 1.$$

Find d from

$$7d \equiv 1 \pmod{120}.$$

Use the Euclidean step

$$120 = 7(17) + 1.$$

Rearrange:

$$1 = 120 - 7(17).$$

The coefficient of 7 is -17 , so

$$d \equiv -17 \pmod{120}.$$

Reduce:

$$-17 + 120 = 103.$$

Therefore

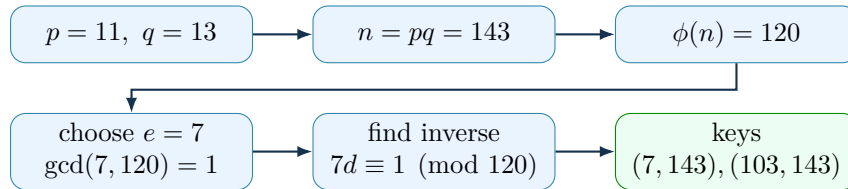
$$d = 103.$$

Answer

Public key = (7, 143)

Private key = (103, 143).

6 Key Generation Trace



7 Why Decryption Works

The private exponent d is chosen so that

$$de \equiv 1 \pmod{\phi(n)}.$$

Therefore there is an integer k such that

$$de = 1 + k\phi(n).$$

For $n = pq$,

$$\phi(n) = (p-1)(q-1),$$

so

$$de = 1 + k(p-1)(q-1).$$

During decryption,

$$C^d \equiv (M^e)^d = M^{ed} \pmod{n}.$$

We need to show

$$M^{ed} \equiv M \pmod{n}.$$

Modulo p

There are two cases.

Case 1: $M \not\equiv 0 \pmod{p}$.

By Fermat's Little Theorem,

$$M^{p-1} \equiv 1 \pmod{p}.$$

Then

$$M^{ed} = M^{1+k(p-1)(q-1)} = M \left(M^{p-1} \right)^{k(q-1)}.$$

Modulo p ,

$$M \left(M^{p-1} \right)^{k(q-1)} \equiv M(1)^{k(q-1)} \equiv M \pmod{p}.$$

Case 2: $M \equiv 0 \pmod{p}$.

Then

$$M^{ed} \equiv 0 \equiv M \pmod{p}.$$

So in all cases,

$$M^{ed} \equiv M \pmod{p}.$$

Modulo q

The same argument gives

$$M^{ed} \equiv M \pmod{q}.$$

Since p and q are coprime, the Chinese Remainder Theorem combines the two congruences:

$$M^{ed} \equiv M \pmod{pq}.$$

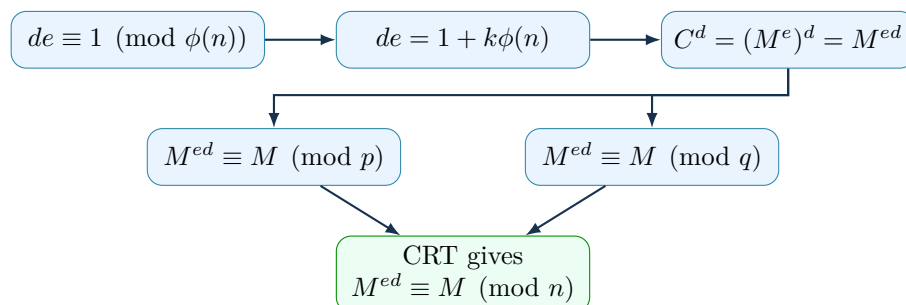
Because $n = pq$,

$$M^{ed} \equiv M \pmod{n}.$$

Conclusion

RSA decryption recovers the original message because d is chosen as the modular inverse of e modulo $\phi(n)$.

8 Correctness Flow



9 Important Notes

Toy Examples Are Not Secure

The examples in this note use tiny primes so the arithmetic is visible. Real RSA uses very large primes and also requires padding schemes in practical cryptographic systems.

Message Range

The message integer must satisfy

$$0 \leq M < n.$$

If a real message is larger than n , it must be encoded and processed in a proper block or padding scheme.

10 Review Box

Core Facts

$$\begin{aligned}n &= pq, & \phi(n) &= (p-1)(q-1). \\ \gcd(e, \phi(n)) &= 1, & de &\equiv 1 \pmod{\phi(n)}. \\ \text{Public key} &= (e, n), & \text{Private key} &= (d, n). \\ C &= M^e \bmod n, & M &= C^d \bmod n. \\ M^{ed} &\equiv M \pmod{n} \text{ explains correct decryption.}\end{aligned}$$